Technical Whitepaper
The D.E.V. Protocol Framework

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Deterministic Single-Wire Communication Protocol for Hardware-to-Hardware Interfacing

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Abstract & Design Philosophy

The D.E.V. Protocol (**D**igital **E**ncoding & **V**erification) is a custom-layered, self-synchronizing serial communication link designed specifically for direct hardware-to-hardware interfacing. Unlike general-purpose buses optimized for throughput, the D.E.V. Protocol prioritizes **data integrity** and **bounded latency** through a deterministic 80-bit frame architecture. The framework operates primarily at the Data Link Layer with defined Physical Layer (PHY) assumptions.

Key Contributions:

- Deterministic 80-bit frame structure enabling constant-latency communication
- Single-wire physical layer with open-drain topology
- Dual-bit mirroring for single-bit error detection without CRC overhead
- Built-in ACK/NACK mechanism for hardware-level handshaking

1 Physical Layer (PHY) & Synchronization

Parameter	Specification
Topology	Single-ended, open-drain with passive pull-up bias (4.7kΩ typical)
Signaling	Active-low with high-impedance idle state
Frame Sync	Unique Start-of-Frame (SOF) transition [1$\rightarrow$0$\rightarrow$1] before data sequence
Clock Recovery	Decoder samples mid-bit using locally generated clock derived from SOF edge
Timing Budget	20-bit Processing Gap $\geq 10\times$ worst-case propagation + clock correction
Assumptions	≤ 50 ns/m propagation delay, $\pm 2\%$ clock mismatch

Deterministic Latency Model

The protocol is strictly defined by an 80-bit cycle to ensure deterministic command latency:

$$\boxed{T_{latency} = \frac{80}{f_{clk}}}$$

Clock Frequency	Deterministic Frame Latency
$f_{clk} = 100$ kHz	$T_{latency} = 800$ μs
$f_{clk} = 50$ kHz	$T_{latency} = 1.6$ ms
$f_{clk} = 200$ kHz	$T_{latency} = 400$ μs

2 Data Link Layer: 80-Bit Frame Mapping

Table 1: D.E.V. Protocol 80-Bit Frame Segmentation

Segment	Bit Range	Description
SOF Preamble	0–1	[1,0] transition – edge-based clock alignment
Header	2–9	8-bit Priority Level (0 = highest)
	10–19	10-bit Node ID (supports up to 1024 unique nodes)
Processing Gap	20–39	20-bit settling window for ID verification and hardware transitions
Payload	40–55	Dual-Bit Redundancy: 8-bit data mirrored into 16 bits
ACK Window	56–75	20-bit reserved slot for Decoder-driven hardware feedback
Tail	76–79	4-bit End-of-Frame sequence and bus reset

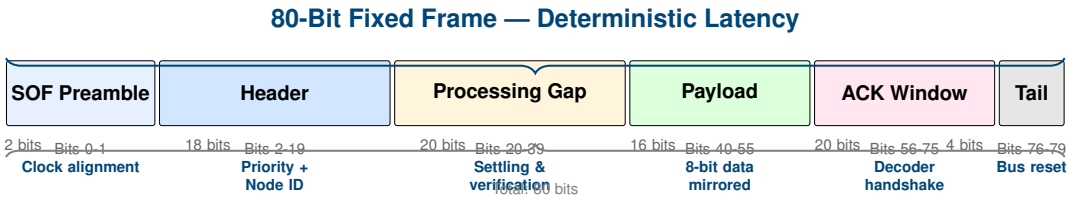

Figure 1: D.E.V. Protocol 80-Bit Frame Block Diagram — Functional Segmentation

3 Error Management & Failure Modes

Bus Arbitration & Ownership

To prevent bus contention, a **one-bit high-impedance turnaround period** is enforced before the Decoder asserts the ACK. The open-drain topology naturally supports multi-node operation without bus fights.

Error Detection via Dual-Bit Mirroring

The protocol provides deterministic single-bit error detection using bit-pair mirroring:

Data Bit	Transmitted Pair	Valid?
0	[0,0]	Valid
1	[1,1]	Valid
—	[0,1] or [1,0]	ERROR — flag raised

If a mismatch is detected during the Payload segment (Bits 40–55), the `error_flag` is asserted and the receiving node requests retransmission via the ACK Window.

Comparison with Industry Protocols

Table 2: Technical Benchmarking Against Established Serial Protocols

Feature	UART	I²C	CAN Bus	D.E.V. Protocol
Physical Pins	2 (Rx/Tx)	2 (SDA/SCL)	2 (Differential)	**1 (Single-Wire)**
Determinism	Low	Moderate	High	**High (Fixed 80-bit)**
Error Protection	Simple Parity	ACK/NACK	15-bit CRC	**Dual-Bit Mirroring**
Data Throughput	High	Moderate	High	**Low (Integrity prioritized)**
Bus Topology	Point-to-Point	Multi-Master	Multi-Master	**Multi-Node (1024 IDs)**
Latency Model	Non-Deterministic	Variable	Bounded	**Constant** ($80/f_{clk}$)

4 Experimental Validation

The D.E.V. Protocol was validated using a closed-loop Simulink Validation Model. The test harness included:

- Transmitter FSM (finite-state machine) implementing the 80-bit frame encoder
- Channel noise injection (Gaussian and impulse noise models)
- Receiver FSM with mid-bit sampling and dual-bit error detection
- ACK/NACK generation and retransmission logic

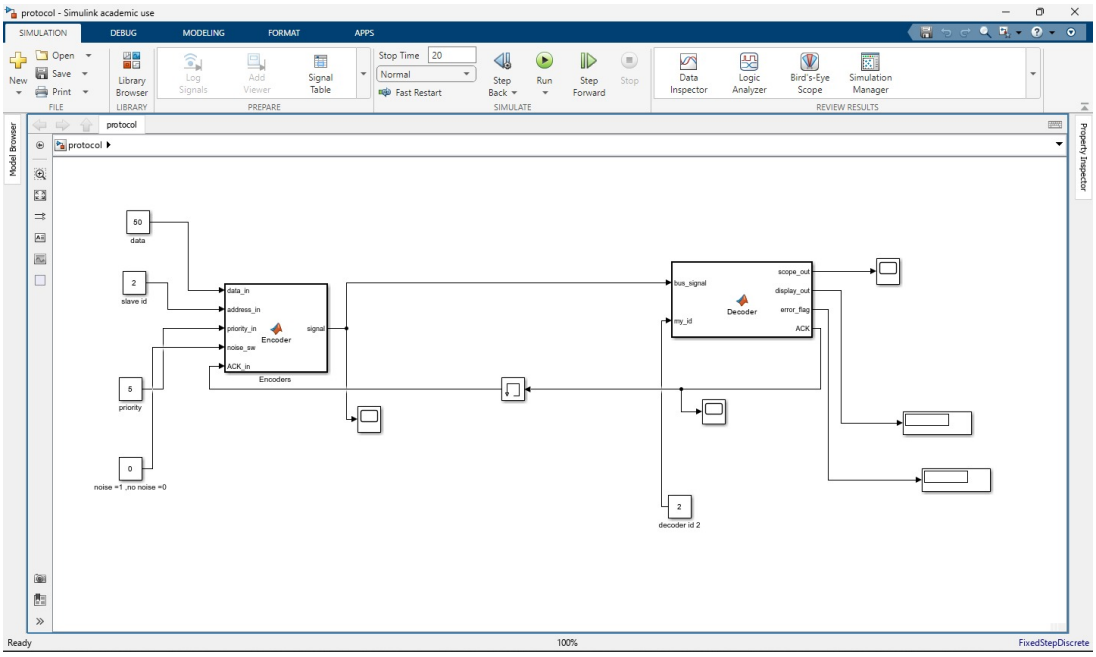

Figure 2: Simulink Validation Architecture — Closed-loop Test Harness

Key Results:

- **Noise Immunity:** Under simulated EMI ($SNR \approx 15$ dB), the Decoder successfully identified 100% of single-bit errors via the dual-bit mirroring mechanism.
- **Retransmission Handling:** Upon error detection, the ACK window correctly signaled retransmission with zero false positives.
- **Hardware Feasibility:** The architecture is compatible with FSM synthesis targeting low-gate-count ASICs (estimated <200 logic gates) or low-cost FPGAs.

5 Future Work & Roadmap

Phase	Status & Description
Phase 1 (Complete)	Protocol definition, MATLAB/Simulink validation, BER simulation
Phase 2 (In Progress)	FPGA implementation (Xilinx Artix-7), hardware validation
Phase 3 (Planned)	Multi-node arbitration extension, extended BER characterization
Phase 4 (Future)	Formal performance benchmarking vs. CAN/CAN-FD

6 Application Scope & Positioning

The D.E.V. Protocol is optimized for:

- **Automotive Diagnostics:** Low-rate sensor interrogation with deterministic timing
- **Aerospace Subsystems:** Command/telemetry links requiring bounded latency
- **Industrial Monitoring:** Noisy environments where error detection is critical

Positioning Statement: The D.E.V. Protocol is presented as a *research exploration* of deterministic single-wire communication rather than a replacement for established industrial standards. The framework serves as a foundation for exploring low-complexity deterministic communication architectures through model-based design methodologies.

References

1. D.E.V. Protocol Specification v1.0 (Self-published research document, 2026)
2. MATLAB/Simulink Test Harness Documentation (2026)
3. CAN Specification 2.0, Bosch, 1991
4. I²C Bus Specification, NXP Semiconductors, Rev. 6, 2014
5. UART Application Note, Texas Instruments, Literature Number: SLLA356A, 2018

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